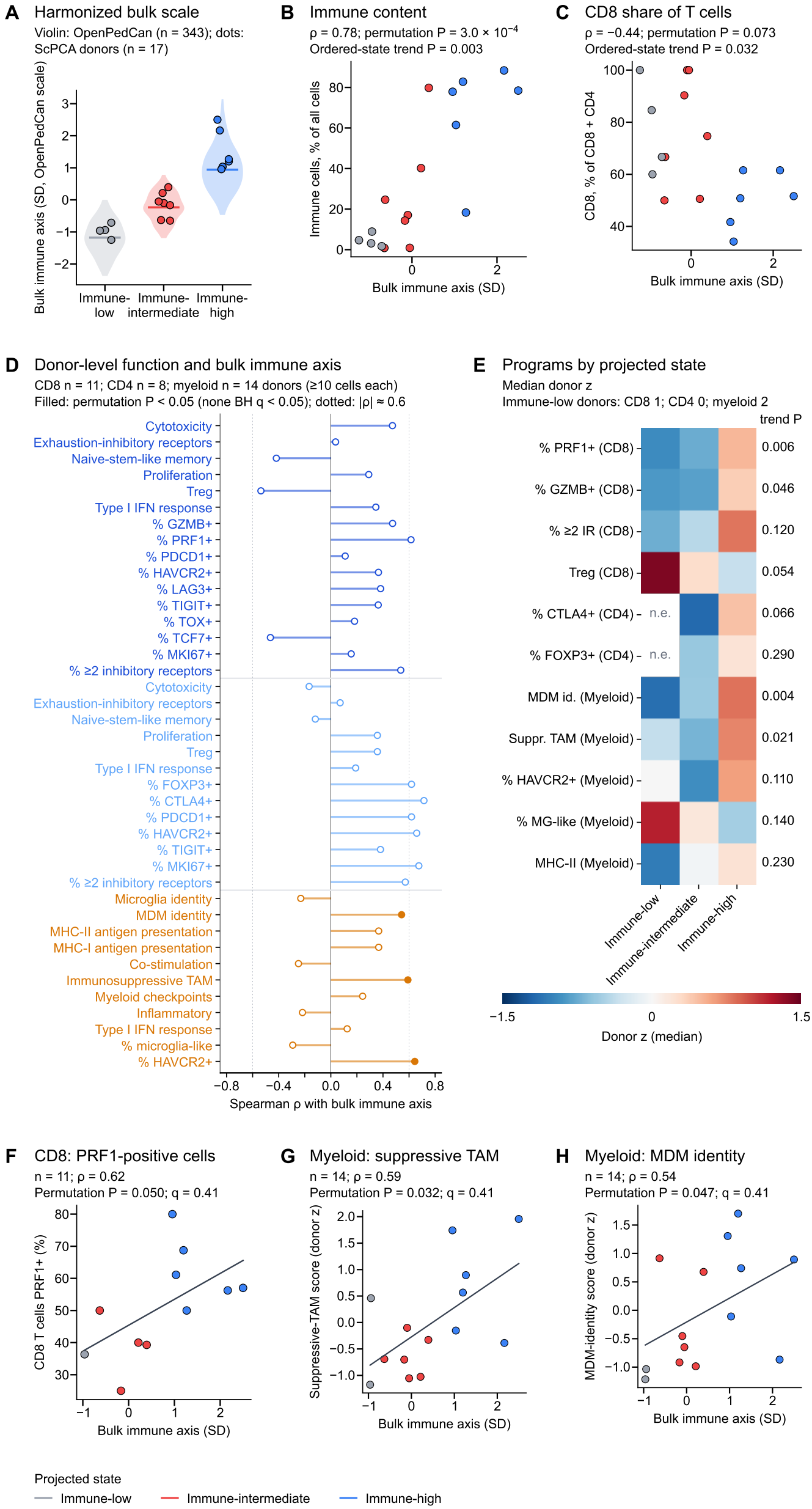

A
Harmonized bulk scale
Violin: OpenPedCan (n = 343); dots: ScPCA donors (n = 17)
Bulk immune axis (SD, OpenPedCan scale)
Immune-low
Immune-intermediate
Immune-high
B
Immune content
ρ = 0.78; permutation P = 3.0 × 10−4
Ordered-state trend P = 0.003
Immune cells, % of all cells
Bulk immune axis (SD)
C
CD8 share of T cells
ρ = −0.44; permutation P = 0.073
Ordered-state trend P = 0.032
CD8, % of CD8 + CD4
Bulk immune axis (SD)
D
Donor-level function and bulk immune axis
CD8 n = 11; CD4 n = 8; myeloid n = 14 donors (≥10 cells each)
Filled: permutation P < 0.05 (none BH q < 0.05); dotted: |ρ| ≈ 0.6
Cytotoxicity
Exhaustion-inhibitory receptors
Naive-stem-like memory
Proliferation
Treg
Type I IFN response
% GZMB+
% PRF1+
% PDCD1+
% HAVCR2+
% LAG3+
% TIGIT+
% TOX+
% TCF7+
% MKI67+
% ≥2 inhibitory receptors
Cytotoxicity
Exhaustion-inhibitory receptors
Naive-stem-like memory
Proliferation
Treg
Type I IFN response
% FOXP3+
% CTLA4+
% PDCD1+
% HAVCR2+
% TIGIT+
% MKI67+
% ≥2 inhibitory receptors
Microglia identity
MDM identity
MHC-II antigen presentation
MHC-I antigen presentation
Co-stimulation
Immunosuppressive TAM
Myeloid checkpoints
Inflammatory
Type I IFN response
% microglia-like
% HAVCR2+
Spearman ρ with bulk immune axis
E
Programs by projected state
Median donor z
Immune-low donors: CD8 1; CD4 0; myeloid 2
trend P
% PRF1+ (CD8)
% GZMB+ (CD8)
% ≥2 IR (CD8)
Treg (CD8)
% CTLA4+ (CD4)
% FOXP3+ (CD4)
MDM id. (Myeloid)
Suppr. TAM (Myeloid)
% HAVCR2+ (Myeloid)
% MG-like (Myeloid)
MHC-II (Myeloid)
n.e.
Immune-low
Immune-intermediate
Immune-high
Donor z (median)
F
CD8: PRF1-positive cells
n = 11; ρ = 0.62
Permutation P = 0.050; q = 0.41
CD8 T cells PRF1+ (%)
Bulk immune axis (SD)
G
Myeloid: suppressive TAM
n = 14; ρ = 0.59
Permutation P = 0.032; q = 0.41
Suppressive-TAM score (donor z)
Bulk immune axis (SD)
H
Myeloid: MDM identity
n = 14; ρ = 0.54
Permutation P = 0.047; q = 0.41
MDM-identity score (donor z)
Bulk immune axis (SD)
Projected state
Immune-low
Immune-intermediate
Immune-high